

Mathematics for Machine Learning

Vazgen Mikayelyan

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WELCOME TO

Mathematics for Machine Learning!

- Course: **Mathematics for Machine Learning**
- Lectures: **Tuesday 18:30-20:30, Saturday 12:00-14:00**
- Instructor: Vazgen Mikayelyan
- Grading: Homework, Midterm, Final Exam

Course Content

- Linear Algebra
- Analytical Geometry
- Mathematical Analysis
- Probability Theory

Find other interesting stuff in the bibliography!

Vectors

Definition

With $n \in \mathbb{N}$ a real-valued n vector a is a "sequence" of n elements a_i , $i = 1, \dots, n$:

$$a = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}, a_i \in \mathbb{R}.$$

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$\mathbb{R}^n$ is the set of all real-valued n -vectors.

Sum of Two Vectors

If a and b are vectors with n elements, then their sum $a + b$ is the following vector:

$$a + b = \begin{bmatrix} a_1 + b_1 \\ a_2 + b_2 \\ \vdots \\ a_n + b_n \end{bmatrix}$$

Multiplication by a Scalar

If a is a vector with n elements and λ is a real scalar, then the scalar multiple λa is the following vector:

$$\lambda a = \begin{bmatrix} \lambda a_1 \\ \lambda a_2 \\ \vdots \\ \lambda a_n \end{bmatrix}$$

- **Distributivity:**

$$(\lambda_1 + \lambda_2)a = \lambda_1 a + \lambda_2 a, \quad a \in \mathbb{R}^n, \lambda_1, \lambda_2 \in \mathbb{R},$$

$$\lambda(a + b) = \lambda a + \lambda b, \quad a, b \in \mathbb{R}^n, \lambda \in \mathbb{R}.$$

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- **Associativity:**

$$(a + b) + c = a + (b + c), \quad a, b, c \in \mathbb{R}^n.$$

Matrices

Definition

With $m, n \in \mathbb{N}$ a real-valued (m, n) matrix A is matrix an $m \cdot n$ -tuple of elements a_{ij} , $i = 1, \dots, m$, $j = 1, \dots, n$, which is ordered in m rows and n columns:

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}, a_{ij} \in \mathbb{R}.$$

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$\mathbb{R}^{m \times n}$ is the set of all real-valued (m, n) -matrices.

Definition

If A and B are $m \times n$ matrices, then the sum $C = A + B$ is an $m \times n$ matrix and the (i, j) entry of the sum is computed as follows:

$$c_{ij} = a_{ij} + b_{ij}.$$

Matrix Addition

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Remark

Shapes of the matrices A and B must be equal.

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$$\forall A, B \in \mathbb{R}^{m \times n} : (A + B) + C = A + (B + C)$$

Matrix Multiplication

Definition

If A is an $m \times n$ matrix and B is an $n \times r$ matrix, then the product $C = AB$ is an $m \times r$ matrix. The (i, j) entry of the product is computed as follows:

$$c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \dots + a_{in}b_{nj} = \sum_{k=1}^n a_{ik}b_{kj}.$$

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Remark

The number of columns of A must be the same as the number of rows of B .

$$c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \dots + a_{in}b_{nj}$$

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{i1} & a_{i2} & \dots & a_{in} \\ \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} \begin{bmatrix} b_{11} & \dots & b_{1j} & \dots & b_{1r} \\ b_{21} & \dots & b_{2j} & \dots & b_{2r} \\ \vdots & & \vdots & & \vdots \\ b_{n1} & \dots & b_{nj} & \dots & b_{nr} \end{bmatrix}$$

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Example

Compute AB if

$$A = \begin{bmatrix} 2 & 3 & 1 \\ 3 & -1 & 0 \end{bmatrix}, \quad \text{and} \quad B = \begin{bmatrix} 1 & -3 \\ 4 & 0 \\ 2 & -2 \end{bmatrix}$$

Check that $AB \neq BA$.

Definition

In $\mathbb{R}^{n \times n}$, we define the **identity matrix**

$$I_n = \begin{bmatrix} 1 & 0 & 0 \dots & 0 \\ 0 & 1 & 0 \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 \dots & 1 \end{bmatrix} \in \mathbb{R}^{n \times n}$$

as the $n \times n$ -matrix containing 1 on the diagonal and 0 everywhere else. With this, $A \cdot I_n = A = I_n \cdot A$ for all $A \in \mathbb{R}^{n \times n}$.

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$$\begin{aligned} \forall A, B \in \mathbb{R}^{m \times n}, C, D \in \mathbb{R}^{n \times p} : \quad & (A + B)C = AC + BC \\ & A(C + D) = AC + AD \end{aligned}$$

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- **Neutral element (identity element):**

$$\forall A \in \mathbb{R}^{m \times n} : \quad I_m A = A I_n = A.$$

Note that $I_m \neq I_n$ for $m \neq n$.

Multiplication by a Scalar

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$$\lambda(BC) = (\lambda B)C = B(\lambda C) = (BC)\lambda, \quad B \in \mathbb{R}^{m \times n}, C \in \mathbb{R}^{n \times k}.$$

Multiplication by a Scalar

Example

Check that

$$(\lambda_1 + \lambda_2)C = \lambda_1 C + \lambda_2 C$$

for any $\lambda_1, \lambda_2 \in R$ and $C := \begin{bmatrix} 1 & 3 \\ 5 & 7 \end{bmatrix}$.

Definition

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Not every matrix A has an inverse A^{-1} . If this inverse does exist, A is called **regular/invertible/non-singular**, otherwise **singular/non-invertible**.

Existence of the Inverse of a 2×2 -Matrix

Consider a matrix

$$A := \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \in \mathbb{R}^{2 \times 2}.$$

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Then for $B = \begin{bmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{bmatrix}$ we get

$$AB = \begin{bmatrix} a_{11}a_{22} - a_{12}a_{21} & 0 \\ 0 & a_{11}a_{22} - a_{12}a_{21} \end{bmatrix} = (a_{11}a_{22} - a_{12}a_{21})I_2.$$

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Therefore

$$A^{-1} = \frac{1}{a_{11}a_{22} - a_{12}a_{21}} \begin{bmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{bmatrix},$$

if $a_{11}a_{22} - a_{12}a_{21} \neq 0$.

Example

The matrices

$$A = \begin{bmatrix} 4 & 5 & 4 \\ 7 & 7 & 6 \\ 2 & 1 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & -1 & 2 \\ 5 & -4 & 4 \\ -7 & 6 & -7 \end{bmatrix}$$

are inverse to each other since $AB = I = BA$.

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Definition

A square matrix A is symmetric if $A = A^T$.

Properties of inverses and transposes

Example

Prove that for any square matrix A the matrices $A + A^T$ and $A \cdot A^T$ are symmetric.

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Prove that for any symmetric matrices A, B the matrix $A + B$ is symmetric.

Remark

The product of two symmetric matrices is not symmetric in general, e.g. the matrices $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, but the matrix $AB = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$ is not symmetric.

Compact Representations of Systems of Linear Equations

The system of linear equations

$$\begin{cases} 2x_1 + 3x_2 - 4x_3 &= 5 \\ x_1 - x_2 + 3x_3 &= -2 \\ -x_1 + 2x_2 + 5x_3 &= 7 \end{cases}$$

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can be written as vector equation

$$\begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix} x_1 + \begin{bmatrix} 3 \\ -1 \\ 2 \end{bmatrix} x_2 + \begin{bmatrix} -4 \\ 3 \\ 5 \end{bmatrix} x_3 = \begin{bmatrix} 5 \\ -2 \\ 7 \end{bmatrix},$$

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and can be compactly written in matrix notation as follows

$$\begin{bmatrix} 2 & 3 & -4 \\ 1 & -1 & 3 \\ -1 & 2 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 5 \\ -2 \\ 7 \end{bmatrix}$$

Compact Representations of Systems of Linear Equations

Note that Ax is a linear combination of the columns of A :

$$\begin{bmatrix} 2 & 3 & -4 \\ 1 & -1 & 3 \\ -1 & 2 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix} x_1 + \begin{bmatrix} 3 \\ -1 \\ 2 \end{bmatrix} x_2 + \begin{bmatrix} -4 \\ 3 \\ 5 \end{bmatrix} x_3$$

Systems of Linear Equations

Let us introduce the general form of a system of linear equations,

$$\begin{aligned}a_{11}x_1 + \dots + a_{1n}x_n &= b_1 \\ \vdots \\ a_{m1}x_1 + \dots + a_{mn}x_n &= b_m\end{aligned}$$

where $a_{ij} \in \mathbb{R}$ and $b_i \in \mathbb{R}$ are known constants and x_j are unknowns, $i = 1, \dots, m$, $j = 1, \dots, n$.

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where $a_{ij} \in \mathbb{R}$ and $b_i \in \mathbb{R}$ are known constants and x_j are unknowns, $i = 1, \dots, m$, $j = 1, \dots, n$. It can be written in matrix form $A\mathbf{x} = \mathbf{b}$, where

$$A = \begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & & \\ a_{m1} & \dots & a_{mn} \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix}$$

Particular and General Solution

Consider the following system of linear equations (SLE):

$$\begin{bmatrix} 1 & 0 & 7 & 2 \\ 0 & 1 & 3 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 5 \\ 8 \end{bmatrix}$$

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or, equivalently $\sum_{i=1}^4 x_i \mathbf{c}_i = \mathbf{b}$, where $\mathbf{c}_i$ is the i th column of the matrix and $\mathbf{b}$ is the right-hand-side.

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Particular and General Solution

All solutions of the SLE, which is called the **general solution**, is the set

$$\left\{ \mathbf{x} \in \mathbb{R}^4; \mathbf{x} = \begin{bmatrix} 5 \\ 8 \\ 0 \\ 0 \end{bmatrix} + \lambda_1 \begin{bmatrix} -7 \\ -3 \\ 1 \\ 0 \end{bmatrix} + \lambda_2 \begin{bmatrix} -2 \\ 4 \\ 0 \\ 1 \end{bmatrix}, \lambda_1, \lambda_2 \in \mathbb{R} \right\}.$$

The algorithm.

Particular and General Solution

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1. Find a particular solution to $A\mathbf{x} = \mathbf{b}$.

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2. Find all solutions to $A\mathbf{x} = \mathbf{0}$.

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1. Find a particular solution to $A\mathbf{x} = \mathbf{b}$.
2. Find all solutions to $A\mathbf{x} = \mathbf{0}$.
3. Combine the solutions from 1 and 2 to the general solution.